

# UNDERSTANDING ATOMIC STRUCTURE

PARTICLES AND FORCES COMBINE TO MAKE UP ATOMS, THE BUILDING BLOCKS OF MATTER

**Solids, liquids, and gases—the three main types of matter—are made up of particles called atoms, each so small that several million would fit side by side on the period at the end of this sentence. All atoms are composed of just three types of particles—protons, neutrons, and electrons.**

In the 19th century scientists became certain of the existence of atoms. The assumption was that atoms were the smallest parts of matter—the word “atom” means indivisible. However, in 1897 the discovery of the electron suggested that atoms have inner structure. Electrons carry negative electric charge, but atoms are neutral overall so a positive charge also had to exist. Scientists used alpha particles, which are produced by radioactive substances, to probe the atom more deeply. In 1911, Ernest Rutherford found that some of the positively charged alpha particles he fired at a thin metal foil bounced back, and proved that this could happen only if each atom has a positive charge concentrated at its center—he had discovered the atomic nucleus.

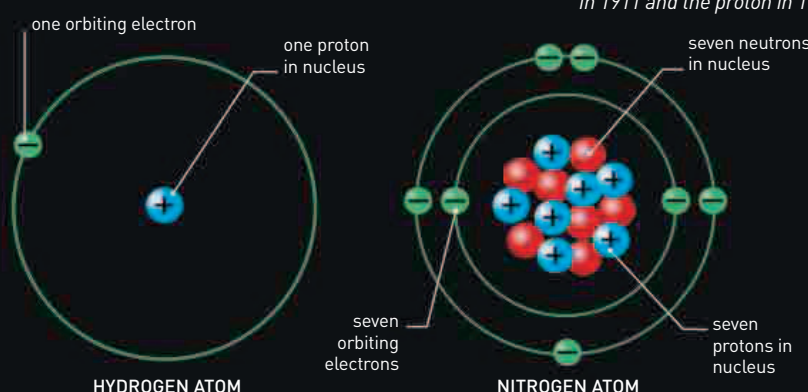
## ATOMS AND ELEMENTS

Except in hydrogen (see right), an atom’s nucleus consists of protons, which are positively charged, and neutrons, which have no charge, held together

by an attractive force called the strong nuclear force. Different elements have different numbers of protons in the nucleus (see below). Negatively charged electrons orbit the nucleus and are prevented from escaping by electric attraction to protons. There are equal numbers of protons and electrons, so atoms have no overall charge.



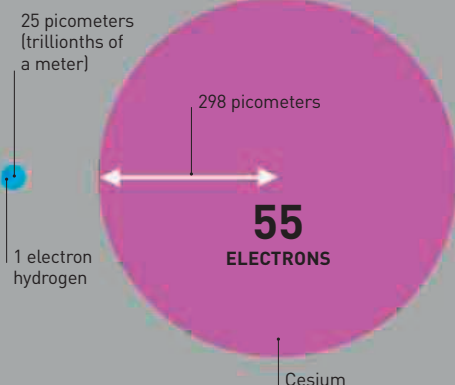
**ERNEST RUTHERFORD**  
New Zealand-born physicist Ernest Rutherford discovered the atomic nucleus in 1911 and the proton in 1920.



## ATOMIC SIZES

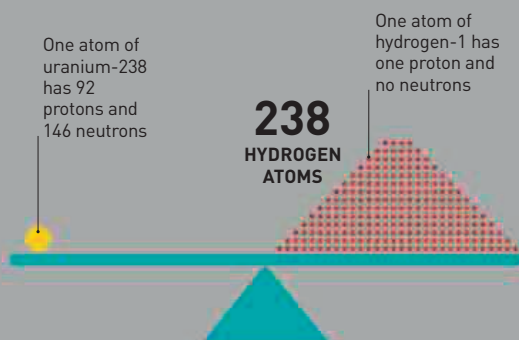
The nucleus accounts for nearly all the mass of an atom but only a tiny proportion of its volume—it is an atom’s electrons that define this. Electrons occupy orbits at specific

distances from the nucleus, and only certain numbers of electrons can occupy each orbit. As a result, atoms with more electrons have more orbits, at increasing distances from the nucleus, producing a larger atomic radius.



## ATOMIC RADIUS

Hydrogen, the lightest element, has only one electron orbiting its nucleus, so it has the smallest atoms. A cesium atom has 55 electrons orbiting its nucleus and is about 12 times wider than a hydrogen atom.



## ATOMIC MASS

Protons and neutrons have identical mass; electrons have negligible mass. So the atomic mass is simply the total number of protons and neutrons. Different versions (isotopes) of an element vary in the number of neutrons.

orbital is a region where there is a high probability of finding electrons

**“ PROTONS  
GIVE AN ATOM  
ITS IDENTITY,  
ELECTRONS  
GIVE IT ITS  
PERSONALITY. ”**

American author Bill Bryson,  
*A Short History of Nearly Everything*, 2003